Natural Deduction

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Slogans

- Deduction system closely resembling natural reasoning
- Focus on *introduction* and *elimination* rules
- Intuitionistic and classical (and minimal) variants
- Interesting completeness proofs
- Connection to algebraic / computational / categorical semantics
- Originally developed by Gentzen in the context of consistency proofs

Overview of Upcoming Lectures on Natural Deduction

Lecture 1:

- Intuitionistic and classical natural deduction
- Structural properties
- Equivalence to sequent calculus
- Double-negation translation

Lecture 2:

- Henkin's completeness proof
- Kripke semantics
- Semantic cut-elimination

Lecture 3:

- Algebraic semantics
- Friedman translation

Intuitionistic Natural Deduction (NJ)

$$\mathsf{Ax}\ \frac{A \in \Gamma}{\Gamma \vdash A} \qquad \mathsf{Exp}\ \frac{\Gamma \vdash \bot}{\Gamma \vdash A} \qquad \mathsf{II}\ \frac{\Gamma, A \vdash B}{\Gamma \vdash A \to B} \qquad \mathsf{IE}\ \frac{\Gamma \vdash A \to B \quad \Gamma \vdash A}{\Gamma \vdash B}$$

$$\mathsf{CI}\ \frac{\Gamma \vdash A \quad \Gamma \vdash B}{\Gamma \vdash A \wedge B} \qquad \mathsf{CE1}\ \frac{\Gamma \vdash A \wedge B}{\Gamma \vdash A} \qquad \mathsf{CE2}\ \frac{\Gamma \vdash A \wedge B}{\Gamma \vdash B}$$

$$\mathsf{DI1}\ \frac{\Gamma \vdash A}{\Gamma \vdash A \vee B} \qquad \mathsf{DI2}\ \frac{\Gamma \vdash B}{\Gamma \vdash A \vee B} \qquad \mathsf{OE}\ \frac{\Gamma \vdash A \vee B \quad \Gamma, A \vdash C \quad \Gamma, B \vdash C}{\Gamma \vdash C}$$

$$\mathsf{AI}\ \frac{\Gamma \vdash A \quad x \notin FV(\Gamma)}{\Gamma \vdash \forall x A} \qquad \mathsf{AE}\ \frac{\Gamma \vdash \forall x A}{\Gamma \vdash A[t/x]}$$

$$\mathsf{EI}\ \frac{\Gamma \vdash A[t/x]}{\Gamma \vdash \exists x A} \qquad \mathsf{EE}\ \frac{\Gamma \vdash \exists x A \quad \Gamma, A \vdash B \quad x \notin FV(\Gamma, B)}{\Gamma \vdash B}$$

Classical Natural Deduction (NK)

To obtain a classical system $\Gamma \vdash_c A$ replace the Exp rule by

$$\mathsf{DN}\frac{\Gamma, \neg A \vdash_c \bot}{\Gamma \vdash_c A}$$

where here we treat $\neg A$ as a shorthand for $A \to \bot$.

We will establish a few expected properties:

- If $\Gamma \vdash A$ then $\Gamma \vdash_c A$.
- We always have $\Gamma \vdash_c A \vee \neg A$.
- We don't always have $\Gamma \vdash A \vee \neg A$.

There is also a minimal system $\Gamma \vdash_m A$ dropping Exp without replacement:

- Obviously, if $\Gamma \vdash_m A$ then $\Gamma \vdash A$ and $\Gamma \vdash_c A$.

Example Derivation: Excluded Middle

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{\neg(A \vee \neg A) \vdash (A \vee \neg A) \to \bot} \quad \dfrac{\dfrac{\dfrac{}{\neg(A \vee \neg A), A \vdash (A \vee \neg A) \to \bot} \quad \dfrac{}{\neg(A \vee \neg A), A \vdash A \vee \neg A}}{\neg(A \vee \neg A), A \vdash \bot}}{\neg(A \vee \neg A) \vdash \neg A}}{\neg(A \vee \neg A) \vdash A \vee \neg A}}{\neg(A \vee \neg A) \vdash \bot}}{\vdash_c A \vee \neg A}$$

Note that already $\neg(A \vee \neg A) \vdash_m \bot$ holds!

Example Derivation: Dual Drinker Paradox

$$\dfrac{\dfrac{\text{previous slide}}{\vdash_c \exists z A(z) \vee \neg\exists z A(z)} \qquad \dfrac{\dfrac{\dfrac{\dfrac{}{\exists z A(z), A(z), \exists y A(y) \vdash_c A(z)}}{\exists z A(z), A(z) \vdash_c \exists y A(y) \to A(z)}}{\exists z A(z), A(z) \vdash_c DD}}{\exists z A(z) \vdash_c DD} \qquad \dfrac{\dfrac{\dfrac{\dfrac{}{\Gamma \vdash_c \exists z A(z) \to \bot} \quad \dfrac{}{\Gamma \vdash_c \exists z A(z)}}{\neg\exists z A(z), \exists z A(z), \neg A(u) \vdash_c \bot}}{\neg\exists z A(z), \exists z A(z) \vdash_c A(u)}}{\neg\exists z A(z) \vdash_c \exists y A(y) \to A(u)}}{\neg\exists z A(z) \vdash_c DD}}{\vdash_c \exists x(\exists y A(y) \to A(x))}$$

Here u is some term, Γ is $\neg\exists z A(z), \exists z A(z), \neg A(u)$ and DD is $\exists x(\exists y A(y) \to A(x))$.

Structural Properties

Recall the structural properties we had for sequent calculus:

- Contraction: if $\Gamma, A, A \vdash \Delta$ then $\Gamma, A \vdash \Delta$
- Exchange: if $\Gamma, A, B \vdash \Delta$ then $\Gamma, B, A \vdash \Delta$
- Weakening: if $\Gamma \vdash \Delta$ then $\Gamma, A \vdash \Delta$

In our formulation of ND based on sets, they all follow from weakening:

Lemma

If $\Gamma \vdash A$ *and* $\Gamma \subseteq \Delta$, *then* $\Delta \vdash A$.

Proof strategy.

Straightforward induction on $\Gamma \vdash A$ for the propositional rules but surprisingly tricky for the quantifier rules due to freshness conditions. □

Direct Consequences of Weakening

- The Exp rule is admissible in NK, so NK subsumes NJ:

$$\dfrac{\dfrac{\Gamma \vdash_c \bot}{\Gamma, \neg A \vdash_c \bot}}{\Gamma \vdash_c A}$$

- The II rule is invertible:

$$\dfrac{\dfrac{\Gamma \vdash A \to B}{\Gamma, A \vdash A \to B} \quad \dfrac{}{\Gamma, A \vdash A}}{\Gamma, A \vdash B}$$

Equivalence of LJ and NJ

Theorem

If $\Gamma \Rightarrow \Delta$ *is derivable in LJ, then* $\Gamma \vdash \bigvee \Delta$ *is derivable in NJ.*

Proof strategy.

By induction on the derivation of $\Gamma \Rightarrow \Delta$, boiling down to showing all rules of LJ admissible in NJ. □

Since we already argued the converse, we obtain:

- NJ is consistent
- Propositional NJ is decidable
- NJ has the disjunction and witness properties
- Completeness results for LJ apply to NJ

Equivalence to LJ and NJ: Interesting Cases

The non-trivial rules to show admissible are:

$$\mathsf{cut}\,\frac{\Gamma_1 \vdash A \qquad \Gamma_2, A \vdash B}{\Gamma_1, \Gamma_2 \vdash B} \qquad \mathsf{RW}\,\frac{\Gamma \vdash \bot}{\Gamma \vdash A}$$

$$\mathsf{L}\neg\,\frac{\Gamma \vdash A}{\Gamma, \neg A \vdash \bot} \qquad \mathsf{L}\to\,\frac{\Gamma_1 \vdash A \qquad \Gamma_2, B \vdash C}{\Gamma_1, \Gamma_2, A \to B \vdash C}$$

$$\mathsf{L}\wedge 1\,\frac{\Gamma, A \vdash C}{\Gamma, A \wedge B \vdash C} \qquad \mathsf{L}\wedge 2\,\frac{\Gamma, B \vdash C}{\Gamma, A \wedge B \vdash C} \qquad \mathsf{L}\vee\,\frac{\Gamma, A \vdash C \qquad \Gamma, B \vdash C}{\Gamma, A \vee B \vdash C}$$

$$\mathsf{L}\forall\,\frac{\Gamma, A[t/x] \vdash B}{\Gamma, \forall x A \vdash B} \qquad \mathsf{L}\exists\,\frac{\Gamma, A \vdash C \qquad x \notin FV(\Gamma, C)}{\Gamma, \exists x A \vdash C}$$

Equivalence of LK and NK

Theorem

If $\Gamma \Rightarrow \Delta$ *is derivable in LK, then* $\Gamma \vdash \bigvee \Delta$ *is derivable in NK.*

Proof strategy.

By induction on the derivation of $\Gamma \Rightarrow \Delta$, boiling down to showing all rules of LK admissible in NK. □

Since we already argued the converse, we obtain:

- NK is consistent
- Propositional NK is decidable
- Completeness results for LK apply to NK

Double-Negation Translation

In propositional logic, the relation of classical and intuitionistic flavour is simple:

Theorem (Glivenko)

If A is propositional, then $\vdash_c A$ if and only if $\vdash \neg\neg A$.

In first-order logic, the situation is slightly more subtle, for instance define:

$$P^* := \neg\neg P \qquad (A \vee B)^* := \neg\neg(A^* \vee B^*)$$
$$(A \to B)^* := A^* \to B^* \qquad (\forall x A)^* := \forall x A^*$$
$$(A \wedge B)^* := A^* \wedge B^* \qquad (\exists x A)^* := \neg\neg\exists x A^*$$

Theorem (Gödel-Gentzen)

For arbitrary A, we have $\vdash_c A$ if and only if $\vdash A^*$.

Consequences: relative consistency/decidability, extraction of computational content...

Double-Negation Translation: Proof Outline

1. Classically, A and A^* are equivalent: $\vdash_c A \Leftrightarrow A^*$
   - Straightforward, exploiting that classical logic identifies B and $\neg\neg B$

2. Double-negation elimination holds for translated formulas: $\vdash \neg\neg A^* \to A^*$
   - By induction on A, exploiting that intuitionistic logic identifies $\neg B$ and $\neg\neg\neg B$

3. Derivations in contexts can be replayed in the translation: $\Gamma \vdash_c A$ implies $\Gamma^* \vdash A^*$
   - By induction on $\Gamma \vdash_c A$, using (2) whenever negative reasoning is needed

Double-Negation Translation: Some Interesting Cases

The most interesting cases are disjunction and existential elimination:

$$\frac{\Gamma^* \vdash \neg\neg(A^* \vee B^*) \qquad \Gamma^*, A^* \vdash C^* \qquad \Gamma^*, B^* \vdash C^*}{\Gamma^* \vdash C^*}$$

$$\frac{\Gamma^* \vdash \neg\neg\exists x A^* \qquad \Gamma^*, A^* \vdash B^* \qquad x \notin FV(\Gamma^*, B^*)}{\Gamma^* \vdash C^*}$$